TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Direct Writing of Physically Unclonable Functions for Smart Part Authentication and Cyber-Physical Security			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Information Security	Service and application security	Information Security	Industrial Security and Convergence Security
	Information & Communication		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Information Security	Common Security Technologies	Plastic Deformation/ Powder	Powder Processing Technologies
Concept of Proposed Technology	**Project overview:** This project aims to develop **one-step, direct-written Physically Unclonable Functions (PUFs) and transform them into secure QR-code digital fingerprints for tamper-resistant smart part authentication and enhanced cyber-physical security.** ➔ Building on the **RPI's patent-pending innovation,** titled *'Blockchain-coupled physical unclonable functions (PUFs) for robust part authentication, traceability, and cyber-physical security"*, this project will lead to high-throughput, scalable fabrication of PUFs embedded directly into the part that securely store mission-critical metadata for intelligent authentication and enhanced cybersecurity (see **Figure 1a**). ➔ As illustrated in **Figure 1b**, we will employ the **Cold Spray (CS) particle deposition** technique [1-4] to deposit optical particles (e.g., fluorescent powders) onto the component surface, keeping features hidden under visible light and revealing them only under UV illumination at specific wavelengths. Owing to the inherently stochastic nature of the CS process—driven by complex two-phase (gas–solid) turbulent flow—the resulting deposition forms highly randomized, spatially distributed particle patterns. Owing to this inherently stochastic nature of the turbulent CS flow [5], the resulting surface structures will be intrinsically unique and difficult to replicate, thereby providing a strong foundation for PUF. ➔ The fabricated **PUFs will be encoded using secure hashing** (e.g., SHA-256 [6-7]) and then algorithmically converted into part-specific, **industry-standard QR codes**.			

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- ➔ Unlike conventional external identifiers (e.g., labels, markings, RFID tags), the proposed approach will **embed the PUFs and its QR-code representation directly into the material surface**, making the security feature an integral part of the component.
- ➔ The proposed technology is **high-throughput, compatible with complex geometries**, and **resilient to demanded environments** (abrasion, temperature, chemical, etc.).
- ➔ The successful realization of this project is expected to **yield a manufacturing-integrated security solution** that shifts trust from external identifiers to the host material (component).

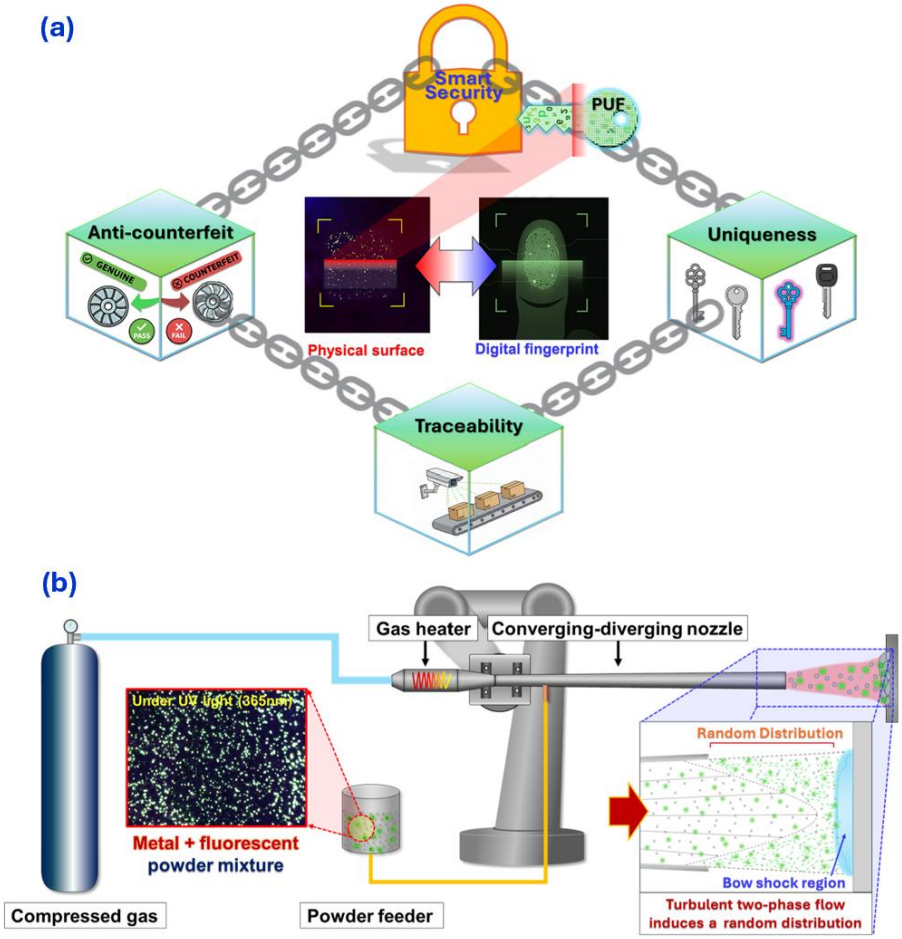

Figure 1: (a) *Schematic illustration of converting a physical surface into a digital fingerprint and its potential applications;* **(b)** *Illustration of the CS direct-write process for PUF fabrication.*

Preliminary results:

Using the robotic Cold Spray (CS) cell in PI Akin's laboratory (**Figure 2**), physically unclonable surfaces were deposited onto commercial service-grade bolts using optical fluorescent particles as the feedstock material (**Figure 3a**). The resulting surfaces remain invisible under standard visible light but becomes clearly distinguishable under UV illumination at 365 nm (**Figure 3b**). Next, the unique surface features were extracted using Voronoi tessellation [8], which partitions the fluorescence pattern into spatially distinct regions (**Figure 3b,** right panel). The surface features include the quartiles of the Voronoi cell areas and cell angles, as shown in **Figure 4**.

Furthermore, PUF metrics, including uniformity and Intra-HD were calculated to validate the uniqueness of the surface across 50 PUF responses. The fabricated surfaces exhibited strong uniqueness, characterized by a mean uniformity of 0.507, Inter-HD of 0.481, and Intra-HD of 0.068, confirming their distinctiveness and suitability for secure, tamper-resistant applications in high-value components.

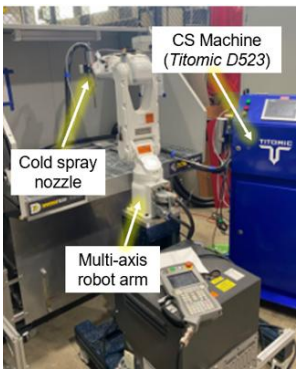

Figure 2. *Robotic CS cell at RPI (in PI Akin's lab).*

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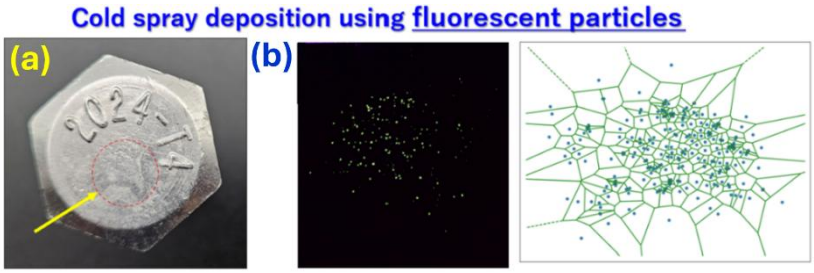
Figure 3. (a) CS-deposited bolt surface; (b) fluorescent particle distribution under UV illumination (left panel) and corresponding Voronoi diagram (right panel).



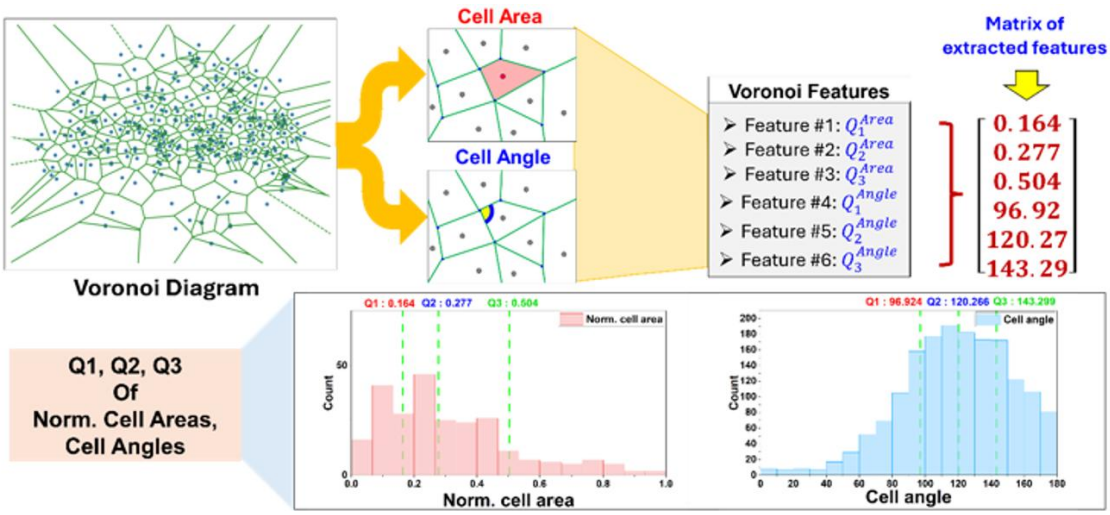


Figure 4. *Voronoi tessellation of the deposited surface for feature extraction to generate PUFs.*

We **hypothesize** that this integrated, full-stack approach—spanning advanced surface fabrication, feature extraction, cryptographic binding, and authentication—will provide robust resistance to tampering, cloning, and counterfeiting in comparison to conventional identification technologies. In collaboration with **Korean industry partners**, the project will assess scalability and technoeconomic feasibility and define pathways toward real-world deployment for a wide range of mission-critical systems.

Potential applications: To mitigate counterfeit parts and enhance cyber-physical security for high-value components across strategic industries, including **aerospace and defense**, **energy infrastructure, advanced manufacturing tooling, medical and industrial devices, and other mission-critical systems**.

Rationale for Joint R&D Project

A joint project is needed for six specific reasons:

- **Need for advanced surface engineering:** While we have demonstrated the feasibility of fabricating PUFs, advancing this capability into an industry-ready technology requires joint expertise in materials processing, surface metrology, and system integration. In particular, the development of durable surfaces that can withstand harsh environmental conditions (e.g., temperature fluctuations, humidity, chemical exposure) demands coordinated R&D efforts to ensure long-term stability, reliability, and performance in real-world applications.
- **Need for stronger anti-counterfeiting defense:** Achieving a truly tamper-resistant solution requires advances in surface engineering and secure cryptographic hashing. A joint R&D effort is therefore necessary to develop next-generation anti-counterfeiting mechanisms that provide robust, multilayered protection against digital threats.
- **Need for intelligent, integrated authentication systems:** The full potential of PUFs can only be realized when paired with intelligent, automated authentication workflows. Advancing this capability requires collaborative efforts to integrate seamless authentication across diverse industrial environments.
- **Need for end-to-end supply-chain traceability:** The proposed approach offers a pathway to robust traceability for high-value components across their entire lifecycle. A joint R&D effort is critical to integrate physically unclonable identifiers with secure digital record-keeping, enabling seamless and tamper-evident tracking in a simulated supply chain.
- **Enhanced cyber-physical security:** Strengthening asset protection in critical

	manufacturing systems requires security solutions that simultaneously safeguard both the physical artifact and its associated digital identity. A collaborative R&D effort will enable the development of integrated cyber-physical security frameworks that combine PUFs, cryptographic linking, and resilient data infrastructures to mitigate spoofing, tampering, and supply-chain infiltration threats. • **Cost and scalability analysis:** Transitioning from laboratory demonstrations to deployable, industry-scale solutions requires a clear understanding of cost structures and integration barriers of the proposed technology. A coordinated R&D effort is therefore needed to assess material and process costs, evaluate manufacturing scalability, and identify pathways for economically viable implementation across diverse industrial environments.
Key Words	1. Cyber-physical security 2. Digital manufacturing security 3. Physically Unclonable Functions (PUFs) 4. Anti-counterfeiting 5. Secure and smart authentication

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	While our preliminary results are promising, an in-depth analysis is required to establish the **direct-write process for PUFs for intelligent part authentication and cyber-physical security**. To this end, the proposed R&D effort will pursue the following objectives: ➔ **Develop surface engineering techniques** to optimize high-throughput fabrication for generating durable PUFs. This includes refining material and process parameters, enhancing resistance to thermal, chemical, and mechanical stressors, and validating long-term stability under real-world operating conditions. ➔ **Develop robust anti-counterfeiting mechanisms** by integrating engineered surface features with secure cryptographic hashing to enable tamper-resistant authentication. This objective focuses on achieving multilayered protection against both physical replication and digital spoofing threats. ➔ **Design intelligent automated authentication workflows** that combine PUF-based identifiers with automated image acquisition, feature extraction, and decision-making algorithms. The goal is to enable seamless, reliable authentication across diverse industrial and operational environments. ➔ **Build an integrated cyber-physical security architecture** that links physical PUF identifiers with resilient cryptographic protocols. This includes evaluating system robustness against spoofing, tampering, and side-channel attacks. ➔ **Establish end-to-end supply-chain traceability framework** that connects PUF-derived identifiers with secure digital records to enable continuous, tamper-evident tracking. This objective includes prototyping data logging, transfer, and verification mechanisms within a simulated or pilot-scale supply chain. ➔ **Perform a techno-economic assessment** to evaluate material costs, cycle-time limitations, equipment requirements, integration barriers, and viable pathways for industrial deployment.			
R&D Target Specifications	The proposed R&D efforts aim to establish quantifiable performance and manufacturing targets to guide the development of direct-written high-performance PUFs. Key target specifications include: • **Durable, High-Throughput PUF Fabrication:** ○ **Thermal stability:** no degradation after **100 cycles** between **−30°C** and **150°C**. ○ **Chemical resistance:** maintain signature fidelity after **24-hour exposure** to mild solvents (IPA, acetone). • **Tamper-Resistant Anti-Counterfeiting Mechanisms** ○ Demonstrate **< 10^{-5}** physical cloning probability via imaging. ○ Maintain **< 2% false-accept rate (FAR)** and **< 1% false-reject rate (FRR)** • **Automated PUF-Based Authentication Workflows** ○ Enable image acquisition and feature extraction in **< 1 min** per authentication event. ○ Achieve **≥ 95% authentication accuracy** across varied lighting and orientation.			

- **Secure Cyber-Physical Integration**
  - Enable TVLA values
  - Mitigation strategies will incur **<2%** area, power and/or latency overheads.
- **Techno-Economic and Deployment Assessment**
  - Target direct-write cycle time of **< 1 min per tag/identifier**.

R&D Plan

To achieve the R&D objectives, we propose a structured task plan as shown in **Figure 5.**

Task 1: Cold-spray direct-writing of physically unclonable surfaces *(University-led)*

In our preliminary studies, polymer-based fluorescent particles were used as optical markers, deposited on surfaces to generate PUFs. However, such polymer materials can degrade or fail under demanding operational conditions, including high temperatures, mechanical stress, and chemical exposure. To address these limitations, this task will systematically investigate the **effects of material selection** and **CS process parameters** on the **quality, durability, and performance of PUFs**. Key variables to be studied include feedstock composition, particle size distribution, carrier gas pressure, substrate and particle temperatures, and nozzle translation speed. The objective is to identify optimal combinations that maximize feature entropy, ensure reproducibility across multiple samples, and enhance adhesion strength on a variety of industrially relevant substrates.

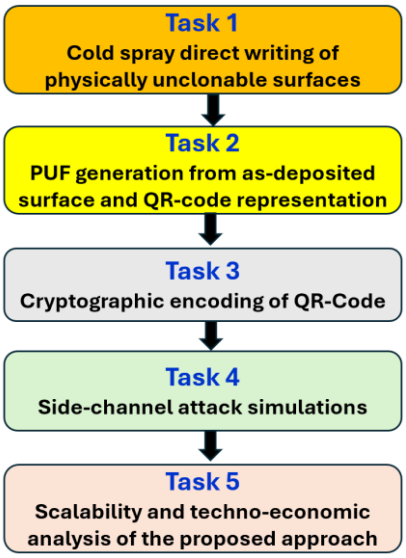


Figure 5. *Proposed workflow.*

Experimental approaches will include: (i) surface feature characterization; (ii) adhesion and mechanical testing; (iii) process-structure correlation; (iv) environmental durability; and (iv) throughput and scalability analysis. The experimental outcomes **will establish robust CS process guidelines for the high-throughput, cost-effective fabrication of durable, high-entropy surface features**. The optimized unclonable surfaces (PUFs) will serve as secure, tamper-resistant identifiers, enabling authentication, anti-counterfeiting, and cyber-physical security applications even under harsh industrial conditions.

Task 2: PUF generation and its QR-code representation *(University-led)*

While PUFs are inherently resistant to cloning and tampering, <u>their digital processing and integration into industrial standards remain a critical challenge</u>. To address this gap, **we will develop an algorithmic approach to convert the generated PUF features into a part-specific QR code**. This representation will enable **smart authentication, end-to-end traceability, and the storage of mission-critical data**. It is important to note that standard QR codes can store approximately 3 kB of data. By leveraging this capability, the PUF-derived information can be securely encoded, digitally linked to the physical part, and easily interrogated using optical scanners for authentication and traceability purposes.

Figure 6 illustrates our proposed approach, demonstrating the algorithmic approach for converting the physical PUF into a digital binary representation. The **hypothesis** is that the binary representation of the PUF can serve as the basis for generating a **part-specific QR code.** This approach will allow the fabricated surface to be digitally captured via PUF responses, enabling secure smart authentication while simultaneously embedding mission-critical information related to the part or process.

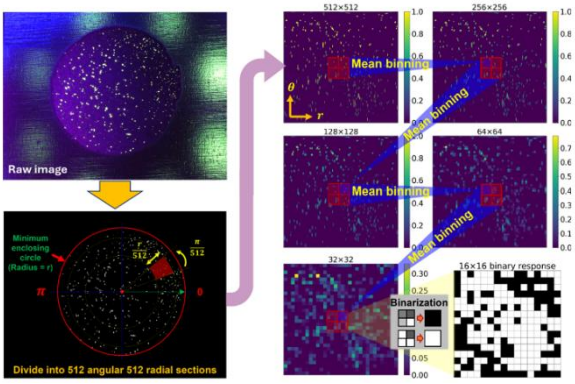


Figure 6. *Converting the physical surface into a binary response for QR code generation.*

To implement this, the QR Coder library in **C#** will be utilized. The binary sequences in will be first converted into hexadecimal format**,** which reduces the data size while preserving all unique features of the PUF. The hexadecimal data is then encoded into the QR code, taking care to maintain sufficient storage capacity for additional metadata, such as product identifiers, manufacturing

parameters, and traceability information. The resulting QR code thus will contain a **compact, secure, and digitally verifiable representation** of the PUF. When scanned, the QR code can be decoded to retrieve the original binary sequence, which can then be compared against the recorded PUF for authentication purposes. This method will ensure both **tamper resistance** and **end-to-end traceability**, allowing the QR code to serve as a bridge between the physical and digital identities of the component.

In addition, we will **develop a graphical user interface (GUI) to automate the authentication process**. Specifically, the designated operator will capture the surface image under pre-specified conditions and upload it through the GUI. The system will then process the image, extract the PUF features, and generate a part-specific QR code automatically. This QR code can be saved and linked to a secure database, enabling seamless authentication and traceability of mission-critical data.

Task 3: Cryptographic encoding of QR-code *(University-led)*

To ensure the security and integrity of PUF-derived QR codes, the binary representations extracted from the physical surfaces will undergo cryptographic processing. Specifically, the binary sequences will be subjected to **secure hash functions** (e.g., SHA-256) to produce a unique digest that serves as a tamper-evident digital signature. The hash, combined with additional metadata, will then be encrypted and embedded within the QR code. This approach is expected to provide multiple security guarantees, including:

(i) **Authentication:** scanned QR codes can be verified by recomputing and comparing the hash to confirm the authenticity of the physical part

(ii) **Data integrity:** Any alteration of the QR code content results in a hash mismatch, enabling immediate detection of tampering

(iii) **Secure metadata embedding:** Manufacturing data, part identifiers, and traceability information can be securely encoded alongside the PUF-derived signature.

Furthermore, we will design a **cryptographic encoding framework for compatibility with industrial digital authentication protocols**, enabling robust cyber-physical security while preserving the uniqueness and high-entropy characteristics of the PUF. In addition, the PIs intend to introduce **Post-Quantum Cryptographic (PQC) algorithms**, like Crystals-Kyber, in the encoding process to protect the proposed method against advanced attacks in the quantum-era. Co-PI Basu will leverage his experiences in this domain [9-10] to develop low-overhead PQC solutions.

Task 4: Side-channel attack simulations *(Joint)*

Side-channel attacks are used to infer confidential information from a computational process, by utilizing physical parameters like power and EM emanations. In the domain of computation, side-channel attacks have been extensively studied over a decade. In the domain of advanced manufacturing, researchers have recently started exploring side-channel attacks. The various physical parameters utilized to this domain are **power, IR-drop, thermal, EM leakage, and optical.** While there are several empirical studies in this domain, to the best of our knowledge, none of these derive a theoretical modeling of the leakage through first principles. Co-PI Basu's team has already performed similar analysis for cryptographic power side-channel leakage and plans to perform the same for advanced manufacturing. To this end, we intend to perform the following tasks:

a) **Develop a theoretical model**, based on first principles, that will be able to replicate the empirical observations for power, EM, optical and thermal side-channel leakage.

b) **Design python-based simulators**, that can accurately model these leakages. These simulators can be used by back-end designers before finalizing on the design of the advanced manufacturing unit.

c) **Develop defensive solutions** based on these simulations, that can be plugged in as "user-defined" modules for securing future advanced manufacturing solutions.

The **PIs envision** that the proposed research will enable a new direction to advanced manufacturing unit design – *i.e.,* consider security as a design parameter Following lab-scale side-channel attack simulations, the performance and robustness of the developed technology will be evaluated **in real-world industrial environments through collaboration with our Korean SME partners**. This approach aims to demonstrate not only the feasibility of secure manufacturing components but also their scalability and resilience in practical applications.

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Task 5: Techno-economic analysis *(Industry-led)*

To transition the PUF-based authentication system from laboratory demonstration to industrial deployment, it is essential to evaluate both scalability and cost-effectiveness. However, its techno-economic viability for industrial deployment is crucial and **guidance from our Korean partners will be vital**. As such, this task will assess the feasibility of production of CS-deposited PUF surfaces and the integration of QR code-based digital authentication into existing manufacturing workflows. To transition the PUF-based authentication system **from laboratory demonstration to industrial deployment**, it is essential to evaluate both scalability and cost-effectiveness. Key objectives include:

- **Material and Operational Cost Analysis:** Evaluate costs associated with feedstock materials, gas usage, energy consumption, equipment operation, and labor requirements.
- **Integration Feasibility**: Analyze compatibility of the CS process and QR code-based authentication with existing industrial systems, including automated surface imaging, data processing, and supply-chain traceability protocols.
- **Techno-Economic Modeling:** Develop a framework to simulate cost, production time, and scalability trade-offs, enabling informed decisions on process optimization and industrial implementation.
- **Risk and Sensitivity Analysis:** Identify critical factors that impact production efficiency, cost, or authentication reliability, and assess potential mitigation strategies.

The outcomes of this task will provide a comprehensive understanding of the economic and operational viability of deploying PUF-based authentication at scale, supporting strategic planning for industrial adoption and commercialization.

3.6. Milestones and Deliverables

The project timelines are summarized in the Table below.

Task	Topic	Year-1	Year-2	Year-3
1	Cold-spray direct-writing of physically unclonable surfaces	----------		
2	PUF generation and its QR-code representation		----------	
3	Cryptographic encoding of QR-code		----------	
4	Side-channel attack simulations		----------	
5	Techno-economic analysis			----------
	Interaction with Korean partners	----------		

Deliverables: The **key deliverables** for this project are:

(1) **Validated cold spray direct-writing platform and PUF fabrication framework.** Specifically; (i) demonstration of high-entropy, durable PUFs produced via optimized CS parameters and (ii) process guidelines for reproducible, single-step PUF generation across multiple substrate types.
(2) **Algorithmic pipeline for PUF-to-binary conversion and QR-code generation:** (i) a complete image-processing workflow converting physical surface features into binary responses; (ii) generation of part-specific QR codes embedding both the PUF signature and metadata; and (iii) a GUI for automated image upload, processing, and QR code output.
(3) **Cryptographically secured QR codes:** Encrypted QR code architecture compatible with industrial authentication systems.
(4) **Side-channel attack simulation and vulnerability assessment**.
(5) **Techno-economic analysis**, including a quantitative evaluation of throughput, cost, and resource requirements for implementing CS-based authentication.

A **comprehensive final report** will also be provided to TIPA, including a detailed summary of project objectives and outcomes; documentation of the CS-enabled PUF fabrication process; established process–structure–performance relationships; the algorithmic pipeline for PUF-to–QR code conversion; an assessment of cyber-physical attack resistance; and preliminary techno-economic analyses. The report will further present risk assessment findings and key recommendations for industrial implementation.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	➔ The principal investigator (**Prof. Semih Akin**) brings expertise in cold spray particle deposition, surface engineering, and advanced manufacturing. He has pioneered the application of cold spray in emerging areas such as flexible and printed electronics, and energy devices [11-13], where solid-state deposition enables the integration of functional materials on temperature-sensitive substrates. ➔ The co-principal investigator (**Prof. Kanad Basu**) brings expertise in hardware security, hardware reliability, and quantum computing. ➔ The co-principal investigator (**Prof. Johnson Samuel**) brings expertise in materials science and advanced manufacturing. ➔ Two Ph.D. students will work on this project. One **Korean student** trained in cold spay by Prof. Akin and Prof. Samuel will perform the experimental and modeling work (Tasks 1-2), while the other student trained by Prof. Basu and Prof. Akin will carry out QR-code generation and side-channel attacks for cyber-physical security evaluations (Tasks 3-4). The techno-economic analysis (Task 5) will be led by the Korean SME partners. ➔ Prof. Akin will be responsible for coordinating the research project. He will schedule weekly meetings for all participants to achieve synergy between computations, material characterizations and electro-chemical testing. He will also provide regular feedback to our Korean partners to ensure that any course corrections can be made in a timely manner. ➔ All the required manufacturing, material synthesis, characterization and manufacturing facilities are available to the investigators on the RPI campus. Specifically, RPI's **Center for Smart Convergent Manufacturing Systems (CSCMS)** provides all required manufacturing capabilities. The **RPI's Materials Research Center (MRC)** are well-equipped for mechanical and microstructural characterization. Together, these facilities provide comprehensive support for the proposed research and ensure seamless collaboration with Korean SME partners.
Expected R&R of Korean SME Partner	The Korean industry partners will advise and assist the RPI team on several critical areas. Expected roles and responsibilities include: ➔ **Technical consultation:** Provide expertise on industrial requirements, manufacturing constraints, and practical deployment considerations relevant to industry sectors. ➔ **Integration into industrial workflows:** Offer guidance on incorporating CS-based PUFs into existing production lines, quality-control protocols, and supply-chain tracking systems. ➔ **Validation and testing support:** Assist in designing and conducting validation experiments under specific industrial operating conditions, including mechanical, thermal, and chemical stress environments. ➔ **Commercial feasibility input:** Provide insights on market adoption challenges, user needs, cost constraints, and scalability expectations within Korean manufacturing ecosystems. ➔ **Feedback on prototypes and deliverables:** Review interim results, prototypes, and documentation to ensure alignment with industry expectations and readiness for pilot testing.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project is expected to establish a **manufacturing innovation** that enables **direct writing of PUFs and their digital conversion into industry-standard QR codes for <u>smart part authentication, end-to-end traceability, and enhanced resistance to cyber-attacks</u>**. The research is expected to deliver significant scientific, technological, and industrial advancements in secure authentication and cyber-physical protection of manufactured components. Key outcomes and their anticipated impacts include: 1. **High-entropy, durable physically unclonable surfaces (PUFs)** capable of withstanding mechanical, thermal, and environmental stressors. 2. **An automated PUF-to-QR smart authentication framework** and user-friendly GUI enabling seamless, scalable integration into manufacturing workflows. 3. **Enhanced cyber-physical security**, combining PUF-derived signatures with cryptographic protection to guard against counterfeiting, spoofing, and supply-chain infiltration. Collectively, these outcomes will lay the foundation for **robust and secure PUF-oriented identifiers** that can be deployed at scale, offering strong potential to improve how high-value components are authenticated, tracked, and trusted across supply chains. The overall workflow will be shared with our industry liaisons in Korea as technology transfer. The PIs will actively work with the liaisons to ensure that the research is integrated into state-of-the-art industrial workflow.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization